

## Scalability & Future Plan Template

|                      |                                         |
|----------------------|-----------------------------------------|
| <b>Date</b>          | 26 June 2026                            |
| <b>Team ID</b>       | SWTID-2026-9014                         |
| <b>Project Name</b>  | AI Powered Advertisement Copy Generator |
| <b>Maximum Marks</b> | 1 Mark                                  |

### Scalability & Future Plan

Describe how the current solution can scale to handle increased load or users, and outline the features or enhancements planned for future releases.

#### Scalability

| Aspect                      | Details                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Current Architecture</b> | Single-server Python Flask application running locally on localhost:5000. Frontend served as static HTML/CSS/JS files via Flask's built-in template engine. AI inference handled externally by Groq API using the Qwen3-32B model. No database — all data processed in-memory per request. Public access provided via Ngrok tunnel                                                                                                                                                                                                                                           |
| <b>Scalability Approach</b> | The application can be scaled horizontally by deploying multiple Flask instances behind a load balancer such as Nginx or AWS Elastic Load Balancer. Gunicorn WSGI server can replace the Flask development server to handle concurrent requests using multiple worker processes. Since the application is stateless — no session data or database — each request is independent, making horizontal scaling straightforward. The Groq API handles all AI inference load externally, meaning the application server itself remains lightweight regardless of prompt complexity |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Expected Growth / Load</b> | The current single-instance Flask setup can handle approximately 10 to 20 concurrent users comfortably. With Gunicorn configured with 4 workers, capacity increases to 40 to 80 concurrent users. Deploying on a cloud platform such as AWS EC2, Google Cloud Run, or Heroku with auto-scaling enabled, the application can scale to handle hundreds or thousands of simultaneous users without architectural changes |
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## Future Plan

| S.No | Planned Feature / Enhancement                                                                                                                                                                                                                                                 | Target Timeline | Priority |
|------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|
| 1    | <b>LinkedIn, X (Twitter), and TikTok Ad Support</b> — Extend the Multi-Platform Adaptation Engine to generate platform-specific copy for LinkedIn sponsored posts, X promoted tweets, and TikTok video ad scripts with platform-specific character limits and tone guidelines | 3 Months        | High     |
| 2    | <b>AI-Powered Image Generation</b> — Integrate an image generation API such as DALL-E or Stable Diffusion to automatically generate advertisement banner visuals alongside the generated copy, creating a complete ready-topublish ad package                                 | 6 Months        | High     |
| 3    | <b>User Authentication and Campaign History Dashboard</b> — Implement user registration and login using Flask-Login with a PostgreSQL database to store past campaigns, allowing users to retrieve, compare, and re-generate previous advertisements                          | 6 Months        | High     |
| 4    | <b>Multi-Language Advertisement Generation</b> — Add support for generating advertisement copy in multiple languages including Hindi, Spanish, French, Arabic, and Mandarin, enabling global marketing campaigns from a single brief.                                         | 9 Months        | Medium   |